Auditing Shortcut Learning and Chemical Out-of-Distribution Generalization in Multimodal Glioblastoma Drug Response Prediction

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Abstract

Predicting anticancer drug sensitivity (LN_IC50) from tumor multi-omics profiles and molecular structures is a core task in computational oncology. On standard in-distribution random splits, multimodal deep learning models frequently report Spearman rank correlations above 0.85. In this work, we investigate whether high correlation on random splits reflects learned drug–gene interactions or a statistical shortcut driven by between-drug potency differences. We construct a leakage-free benchmark for Glioblastoma (GBM) combining GDSC2 dose-response assays, DepMap transcriptomics, somatic hotspot mutations, promoter methylation, and RDKit molecular fingerprints across $5,107$ unique $(\text{drug}, \text{cell line})$ pairs (24 GBM cell lines and 243 drugs). We evaluate a Two-Tower Multi-Layer Perceptron (`BaselineMLP`) implementing Joint (Intermediate) Fusion against closed-form statistical baselines across three evaluation regimes: Random (in-distribution), Leave-Cell-Out (unseen patient cell lines), and Bemis–Murcko Scaffold (unseen chemical structures) across 5 random seeds. First, on the Random split, an interaction-free additive baseline ($\bar{y}_d + \bar{y}_c - \bar{y}_{\text{global}}$) achieves a Spearman correlation of 0.8752, matching `BaselineMLP` (0.8744 ± 0.0051) and showing that 95.5% of the explainable rank correlation comes from memorizing average drug toxicity (`Drug-Only` $r_s = 0.8349$). Second, when evaluating on unseen chemical scaffolds, where drug-identity lookup is impossible, global Spearman drops to 0.2905 ± 0.0045 and RMSE increases from 1.0346 to 2.1951 due to severe prediction variance shrinkage toward the training mean ($\sigma(\hat{y})/\sigma(y) = 0.663$). Third, despite global collapse on unseen scaffolds, the median within-drug Spearman correlation across the 24 GBM cell lines remains 0.530 (compared to 0.500 on Random and Leave-Cell-Out). This decoupling shows that the multi-omics cell encoder preserves relative patient susceptibility ranking for novel compounds, whereas the primary bottleneck in chemical out-of-distribution generalization is calibrating the per-drug baseline potency offset.

1 Problem formulation

1.1 Clinical motivation in Glioblastoma

Glioblastoma (GBM, World Health Organization Grade IV astrocytoma) is the most common and aggressive primary malignant brain tumor in adults. Under the standard Stupp protocol—maximal

*Course Final Mini-Project for *MD6329: Deep Learning in Healthcare AI*, Nanyang Technological University (NTU). Project Showcase & Interactive LecTrace Demo: https://mnemopolis.github.io/gbm_drug_response_project/. Source Code & Reproducible Pipeline: https://github.com/mnemopolis/gbm_drug_response_project.

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safe surgical resection followed by concomitant radiotherapy and oral chemotherapy with the DNA-alkylating agent Temozolomide (TMZ)—median overall survival remains under 15 months, and five-year survival is below 7% [1]. Clinical response to TMZ and salvage therapies (such as Lomustine, Carmustine, Irinotecan, and Etoposide) varies widely across patients due to intratumoral and interpatient molecular heterogeneity [3]. A key clinical biomarker in GBM is epigenetic silencing of the O^6-methylguanine-DNA methyltransferase (`MGMT`) gene via promoter methylation: tumors with low `MGMT` expression cannot efficiently repair O^6-methylguanine DNA lesions induced by TMZ and are more sensitive to alkylating chemotherapy, whereas `MGMT`-high tumors exhibit resistance [2]. Together with recurrent somatic alterations in `PTEN`, `TP53`, `EGFR`, and `RB1`, these molecular features motivate computational models that predict drug sensitivity from multi-omics profiles.

1.2 The random-split evaluation pitfall and research questions

Preclinical drug response models are typically trained on high-throughput cell-line screening panels such as the Genomics of Drug Sensitivity in Cancer (GDSC2) [4, 5], where sensitivity is quantified by the natural logarithm of the half-maximal inhibitory concentration (LN_IC50, in $\ln\ \mu$M). Most published deep learning studies evaluate models using an in-distribution *random pair split*, holding out 20% of $(\text{drug}, \text{cell line})$ pairs at random. Under this setup, the same 243 drugs and 24 GBM cell lines appear in both the training and test sets. Because screened compounds span a wide range of intrinsic potencies—from general cytotoxic agents such as Docetaxel (mean LN_IC50 $= -4.12$) to weak in-vitro alkylators such as Temozolomide (mean LN_IC50 $= +6.45$)—between-drug variance is much larger than between-cell-line variance within a given drug [19]. Consequently, a neural network can achieve a high pooled correlation on a random split simply by memorizing each drug's mean LN_IC50, without learning how specific transcriptomic or mutational profiles modulate drug response.

To test whether a multimodal neural network learns biological drug–cell interactions or relies on drug-mean memorization, we formulate three research questions:

- **RQ1 (Shortcut audit on in-distribution splits):** How much of the test Spearman rank correlation achieved by a Two-Tower multimodal neural network (`BaselineMLP`) on a random split exceeds a zero-interaction additive baseline ($\hat{y}_{d,c} = \bar{y}_d + \bar{y}_c - \bar{y}_{\text{global}}$) that sums marginal drug and cell-line training means?
- **RQ2 (Biological vs. chemical out-of-distribution shift):** How does predictive performance degrade when generalizing to unseen GBM patient cell lines (*Leave-Cell-Out split*) versus unseen Bemis–Murcko chemical scaffolds (*Scaffold split*), and how does prediction variance shrinkage ($\sigma(\hat{y})/\sigma(y)$) behave across these regimes?
- **RQ3 (Decoupling between global and within-drug ranking):** When global pooled correlation drops on unseen chemical scaffolds, does the model lose all predictive signal, or does the multi-omics cell encoder still rank relative cell-line susceptibility *within* each novel drug?

Hypothesis and study plan. We hypothesize that (i) random-split performance in GBM drug response prediction is dominated by drug-level main effects with negligible nonlinear interaction gain over an additive baseline, (ii) Leave-Cell-Out performance remains high because known drug mean potencies transfer across GBM cell lines, and (iii) Scaffold splitting removes the drug-identity shortcut, causing predictions to shrink toward the global training mean and degrading global correlation while preserving within-drug relative patient rankings.

1.3 Role of Software 3.0 (Generative AI copilots) in problem formulation

During initial scoping, we used DeepSeek for early project ideation, problem conceptualization, and modular system architecture design, alongside Gemini for baseline prototyping and code implementation. Initial AI suggestions proposed replacing fingerprint encoders with deeper graph attention networks and cross-modal transformers to push random-split Spearman correlation from 0.87 toward 0.91. When we inspected the empirical LN_IC50 distributions, however, we observed that drug-level means spanned over 10 log units whereas cell-line means varied by less than 1.5 log units. We prompted the AI copilots to help implement closed-form marginal baselines (`Cell-Only`, `Drug-Only`, and `Additive`), which revealed that the zero-interaction additive baseline matched the neural network on the random split (0.8752 vs. 0.8744). This check led us to reject the initial

suggestion of chasing higher random-split accuracy with heavier architectures, and instead focus on out-of-distribution (OOD) auditing, variance shrinkage, and metric decoupling.

2 Related work

Pharmacogenomic screens and multimodal drug response models. Large-scale pharmacogenomic repositories, including GDSC [4, 5] and CCLE / DepMap [6–8], have enabled data-driven modeling of cancer drug sensitivity. Early approaches used elastic net regression, random forests, or kernel methods over gene expression profiles [5, 9]. Subsequent deep learning models introduced two-tower intermediate/joint-fusion architectures [21] that separately encode drug structures and cellular multi-omics features before concatenating latent representations to predict IC_{50}. Representative models include CDRscan [10], which applies convolutional layers to mutation profiles and molecular fingerprints; DeepCDR [11], which combines a hybrid graph convolutional network with multi-omics subnetworks for transcriptomics, genomics, and epigenomics; GraphDRP [12], which benchmarks GIN and GCN molecular graph encoders against 1D convolutional omics encoders; and TGSA [13], which incorporates protein–protein interaction priors into cell-line graphs. While these studies report strong correlations ($r_s \in [0.85, 0.92]$) on random pair splits, most do not quantify how much of that correlation is already explained by marginal drug averages.

Molecular representations and chemical scaffold splitting. Extended-Connectivity Fingerprints (ECFPs / Morgan circular fingerprints) [14] in RDKit [24] hash circular atomic neighborhoods of radius r into fixed-length bit vectors. For evaluating chemical generalization, Bemis and Murcko [15] decomposed drug molecules into ring systems, linkers, and side chains, and MoleculeNet [16] established Bemis–Murcko scaffold splitting as a standard benchmark for testing generalization to novel chemical series. On molecular graphs, Xu et al. [17] showed that Graph Isomorphism Networks (GIN) match the discriminative power of the Weisfeiler–Lehman isomorphism test, motivating our inclusion of both a Morgan fingerprint MLP tower and a GIN graph encoder (`GNNDrugMLP`).

Shortcut learning and cold-start auditing in biomedical AI. Geirhos et al. [18] characterized *shortcut learning* as the reliance of neural networks on simple marginal regularities that succeed on i.i.d. splits but fail under distribution shift. In pharmacogenomics, Li et al. [19] and Sharifi-Noghabi et al. [20] showed that pooling multiple drugs into a single correlation metric inflates apparent performance when between-drug variance dominates within-drug variance, and advocated strict cell-line cold-start (Leave-Cell-Out) and drug cold-start (Scaffold) evaluation. We extend these principles to GBM by jointly auditing additive baseline decomposition, multi-seed stability, prediction variance shrinkage ($\sigma(\hat{y})/\sigma(y)$), and the decoupling between global correlation and within-drug patient stratification.

Connections to MD6329 course lectures. Our study design and experimental audit directly operationalize four core modules from the *MD6329: Deep Learning in Healthcare AI* lecture series[2]:

- **Entity-disjoint splitting and leakage-free standardization (Lecture 02, Slides 73–81, 107, 123–124; Lectures 05–06):** Lecture 02 (*Slide 107 & Slides 123–124: "What Counts as a Data Example?"*) demonstrates that correlated measurements from the same biological entity (e.g., 200 slices from one patient CT volume) are not independent examples and must never be split randomly across train and test sets. Analogously, our $5,107$ GBM assays derive from only 24 cell lines and 243 drugs; naive random pair splitting leaks drug and cell identities, motivating our entity-disjoint *Leave-Cell-Out* and *Bemis–Murcko Scaffold* splits (`src/splitting.py`). Following *Lecture 02, Slides 73–81* (holding out test sets for final unbiased evaluation and fitting zero-mean, unit-variance standardization on training data) and *Lecture 05, Slide 79 / Lecture 06, Slides 15–20* (missing-value imputation), `src/standardization.py` fits `StandardScaler`, PCA, and `MGMT` median imputation strictly on unique training cell lines.
- **Multimodal representation learning and Joint Fusion—Type I (Lecture 08, Slides 3, 9–17, 29–33; Lecture 10, Slides 8, 14–15):** Lecture 08 (*Slide 3*) distinguishes *Very Different Modalities* with heterogeneous dimensionality and structure—matching our pairing of

[2]MD6329 Course Lecture Recordings Playlist: https://www.youtube.com/playlist?list=PLORG7zBQosxw.

Table 1: Summary of the deduplicated GBM cohort and the three evaluation partitions (15% of the training partition is held out for validation early stopping).

Split Regime	Train Pairs	Val Pairs	Test Pairs	Train Cells	Test Cells	Test Drugs	Disjointness Constraint
Random (In-Dist.)	3,472	613	1,022	24	24	236	Pair overlap = 0
Leave-Cell-Out (OOD)	3,611	638	858	20	4	243	Cell `ModelID` overlap = 0
Scaffold (Chem. OOD)	3,447	609	1,051	24	24	47	Murcko scaffold overlap = 0

sparse 2048-bit binary Morgan fingerprints with continuous cellular multi-omics. Under the multimodal fusion taxonomy of Huang et al. [21] presented in *Lecture 08, Slides 29–33*, concatenating raw inputs (*Early Fusion, Slide 31*) lets the 2048-D fingerprint overwhelm the 44-D omics vector, whereas *Late Fusion (Slide 33)* aggregates separate predictions only at the output stage. Our Two-Tower `BaselineMLP` implements *Joint (Intermediate) Fusion—Type I (Slides 9, 14–15, 29, 32)*: modality-specific encoders (`drug_net`: $2048 \to 512 \to 256$; `cell_net`: $d_{\text{cell}} \to 128 \to 256$) project each input into balanced $\mathbb{R}^{256}$ intermediate representations before concatenation (512-D) and shared prediction layers, with MSE loss gradients backpropagating end-to-end. Moreover, compressing $19,138$ genes into K train-only PCs ($\dim(z) \ll \dim(x)$) directly applies the dimensionality-reduction principle of *Lecture 10, Slides 8, 14–15*.

- **Distribution shift robustness and Model Cards for responsible clinical AI (Lecture 11, Slides 40–42, 52–55):** Lecture 11 (*Slides 40–42*) examines model degradation under clinical distribution shift [22] and worst-group performance, which we quantify under biological OOD (`leave_cell`) and chemical OOD (`scaffold`), revealing severe variance shrinkage ($\sigma(\hat{y})/\sigma(y) = 0.663$). Finally, following the *Model Cards for Model Reporting* framework [23] covered in *Lecture 11, Slides 52–54*, we explicitly delineate valid clinical scope (relative patient biomarker stratification via preserved within-drug ranking, $r_s = 0.530$) from prohibited out-of-scope deployment (uncalibrated clinical dosing for novel scaffolds).

3 Data

3.1 Data sources, replicate deduplication, and cohort statistics

We integrate five public tables from GDSC2 [4, 5], DepMap / CCLE [7, 8], and PubChem (`src/acquisition.py` and `src/fingerprints.py`): (1) **GDSC2 dose-response records** (`GDSC2_fitted_dose_response_27Oct23.csv`), filtered for `CANCER_TYPE` containing "Glioblastoma" and mapped to DepMap `ModelID` (`ACH-*`) via `Model.csv` (8,603 raw records); (2) **Transcriptomic gene expression** (`OmicsExpressionProteinCodingGenes...csv`), providing batch-corrected $\log_2(\text{TPM} + 1)$ values across 19,138 protein-coding genes plus the explicit `MGMT` expression biomarker (`MGMT_Expr`); (3) **Somatic hotspot mutations** (`OmicsSomaticMutationsMatrixHotspot.csv`), retaining 20 active driver genes with non-zero mutation frequency across the GBM cohort (including `TP53`, `PTEN`, `EGFR`, `RB1`, `NF1`, and `PIK3CA`); (4) **Promoter DNA methylation** (`CCLE_RRBS_TSS_1kb_20180614.txt`), providing RRBS TSS 1 kb methylation to cross-check epigenetic `MGMT` silencing; and (5) **Chemical structures** (`drug_smiles.csv`), converted via RDKit [24] into 2048-bit radius-2 Morgan fingerprints ($\mathbf{f}_d \in \{0,1\}^{2048}$) and Bemis–Murcko scaffold SMILES strings.

In raw GDSC2, compounds are often assayed multiple times against the same cell line across screening plates (8,603 raw records vs. 5,107 unique pairs). Splitting raw records without deduplication places technical replicates of the same (drug, cell line) pair into both train and test sets. In `src/deduplication.py`, we average LN_IC50 across replicates for each unique (ModelID, DRUG_NAME) pair and normalize drug names (`normalize_drug_name`) prior to alignment. The resulting cohort comprises 5,107 unique pairs across 24 GBM/Gliosarcoma cell lines and 243 drugs (189 unique Murcko scaffolds, 0.0% missingness in `MGMT_Expr`), with LN_IC50 mean 2.85 $\ln \mu$M and standard deviation 2.58 $\ln \mu$M (range $[-6.75, +11.23]$; Table 1 and Figure 1).

Qualitative biological and pharmacological examples. The dataset reflects key clinical contrasts in GBM: cell line `U-87 MG` (`ACH-000075`) has silenced/low `MGMT` expression (MGMT_Expr = 0.420) with a `PTEN` mutation, whereas `T98G` (`ACH-000571`) has high `MGMT` expression (MGMT_Expr = 5.840) and dual `PTEN`/`TP53` mutations, conferring TMZ resistance. Across drugs,

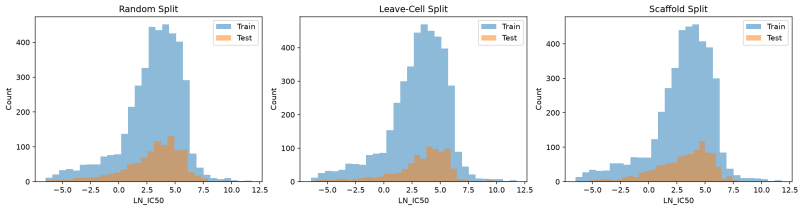


Figure 1: Train and test LN_IC50 label distributions across the Random, Leave-Cell-Out, and Bemis–Murcko Scaffold splits ($5,107$ total pairs across 24 GBM cell lines and 243 drugs).

standard oral alkylator Temozolomide (scaffold `O=c1ncc2ncn([H])c2n1`) has a high mean in-vitro LN_IC50 of $+6.45$, Irinotecan has $+2.76$, Carmustine (BCNU) has $+5.63$, and microtubule inhibitor Docetaxel has -4.12.

4 Approach

4.1 Leakage-free omics standardization and automated pipeline auditing

Applying standardization or PCA across all cell lines before splitting leaks test-set gene covariance into training features. In `src/standardization.py` (`fit_transform_cell_pca_features`), we fit all transformations strictly on unique training cell lines $\mathcal{C}_{\text{train}}$: (1) a per-gene Z-score `StandardScaler` ($\boldsymbol{\mu}_{\text{train}}, \boldsymbol{\sigma}_{\text{train}}$) is fitted only on $\mathbf{X}^{\text{train}}_{\text{expr}} \in \mathbb{R}^{|\mathcal{C}_{\text{train}}| \times 19{,}138}$; (2) PCA is fitted on the scaled training cells with $K = \min(32, |\mathcal{C}_{\text{train}}| - 1)$ components, yielding $K = 23$ on Random/Scaffold (24 training cells) and $K = 19$ on Leave-Cell-Out (20 training cells), and test cells $\mathcal{C}_{\text{test}}$ are projected onto the frozen axes; (3) concatenating the K PCA coordinates, 20 binary hotspot mutations, and training-median-imputed `MGMT_Expr` yields $\mathbf{x}_c \in \mathbb{R}^{d_{\text{cell}}}$ with $d_{\text{cell}} = 44$ (Random/Scaffold) or $d_{\text{cell}} = 40$ (Leave-Cell-Out). After splitting, `src/pipeline_audit.py` (`run_full_pipeline_audit`) programmatically asserts zero pair overlap, zero cell overlap on `leave_cell`, zero scaffold overlap on `scaffold`, removal of raw `EXPR_*` columns, and zero NaNs.

4.2 Closed-form statistical and chemical similarity baselines

To separate marginal main effects from learned drug–cell interactions, `src/baselines.py` defines three closed-form statistical baselines from training means ($\bar{y}_{d,\cdot}$ for drug d, $\bar{y}_{\cdot,c}$ for cell line c, and global mean $\bar{y}_{\text{train}}$):

$$\hat{y}^{\text{cell}}_{d,c} = \bar{y}_{\cdot,c}, \qquad \hat{y}^{\text{drug}}_{d,c} = \bar{y}_{d,\cdot}, \qquad \hat{y}^{\text{add}}_{d,c} = \bar{y}_{d,\cdot} + \bar{y}_{\cdot,c} - \bar{y}_{\text{train}}. \tag{1}$$

On Leave-Cell-Out, $\bar{y}_{\cdot,c}$ has 0% test coverage; on Scaffold, $\bar{y}_{d,\cdot}$ has 0% test coverage. For unseen drugs d' on the Scaffold split, we also evaluate a **Nearest-Drug Morgan FP Baseline** (`evaluate_nearest_drug_scaffold_baseline`), which predicts $\hat{y}^{\text{NN}}_{d',c} = \bar{y}_{d^*(d'),\cdot}$ using the training drug $d^*(d') = \arg\max_{d \in \mathcal{D}_{\text{train}}} \frac{\mathbf{f}_{d'}^\top \mathbf{f}_d}{\|\mathbf{f}_{d'}\|_2 \|\mathbf{f}_d\|_2}$ with highest Morgan fingerprint cosine similarity.

4.3 Two-Tower multimodal neural network architecture and training

Our primary neural architecture is the Two-Tower Multi-Layer Perceptron (`BaselineMLP` in `src/mlp_model.py`), which implements **Joint (Intermediate) Fusion—Type I** (*MD6329 Lecture 08, Slide 32* [21]) by encoding drug Morgan fingerprints $\mathbf{f}_d \in \mathbb{R}^{2048}$ and cell multi-omics vectors $\mathbf{x}_c \in \mathbb{R}^{d_{\text{cell}}}$ into $\mathbb{R}^{256}$ intermediate representations before joint concatenation:

$$\mathbf{h}_{\text{drug}} = \mathbf{W}_{d,2}\,\text{Dropout}_{0.3}(\text{ReLU}(\mathbf{W}_{d,1}\mathbf{f}_d + \mathbf{b}_{d,1})) + \mathbf{b}_{d,2} \in \mathbb{R}^{256}, \quad (2048 \to 512 \to 256), \tag{2}$$

$$\mathbf{h}_{\text{cell}} = \mathbf{W}_{c,2}\,\text{Dropout}_{0.3}(\text{ReLU}(\mathbf{W}_{c,1}\mathbf{x}_c + \mathbf{b}_{c,1})) + \mathbf{b}_{c,2} \in \mathbb{R}^{256}, \quad (d_{\text{cell}} \to 128 \to 256), \tag{3}$$

$$\hat{z}_{d,c} = \mathbf{w}_{\text{out}}^\top\,\text{Dropout}_{0.3}(\text{ReLU}(\mathbf{W}_{\text{fuse}}[\mathbf{h}_{\text{drug}} \,\|\, \mathbf{h}_{\text{cell}}] + \mathbf{b}_{\text{fuse}})) + b_{\text{out}} \in \mathbb{R}, \quad (512 \to 128 \to 1). \tag{4}$$

We also implement an exploratory Graph Isomorphism Network (`GNNDrugMLP`) using two `GINConv` layers ($16 \to 128 \to 256$) with global mean pooling over SMILES molecular graphs (`src/dataset.py`).

Table 2: Benchmark comparison of statistical baselines and Two-Tower `BaselineMLP` (5 random seeds, mean ± SD) across the three evaluation regimes (5,107 GBM pairs). "N/A" indicates 0% test coverage due to unseen cell lines or unseen drug scaffolds.

Split Regime	Cell-Only r_s	Drug-Only r_s	Additive r_s	Nearest-FP r_s	`BaselineMLP` Global r_s	`BaselineMLP` RMSE	Within-Cell MLP r_s	Within-Drug Med. r_s
Random (In-Dist.)	0.2107	0.8349	**0.8752**	—	0.8744 ± 0.0051	$\mathbf{1.0346 \pm 0.0211}$	0.8541 ± 0.0576	0.500
Leave-Cell-Out (All 5)	N/A	0.8362	N/A	—	0.6947 ± 0.2979	1.3970 ± 0.6115	$\mathbf{0.8460 \pm 0.0608}$	0.500
Leave-Cell-Out (4 Stable)*	N/A	0.8362	N/A	—	$\mathbf{0.8436 \pm 0.0034}$	$\mathbf{1.0912 \pm 0.0085}$	$\mathbf{0.8460 \pm 0.0608}$	0.500
Scaffold (Chem. OOD)	0.2494	N/A	N/A	0.1445	$\mathbf{0.2905 \pm 0.0045}$	2.1951 ± 0.0081	0.1735 ± 0.0813	**0.530**

*On Leave-Cell-Out, seeds $0, 1, 3, 4$ converge stably ($r_s \in [0.8388, 0.8476]$), whereas seed 2 early-stops at epoch 21 ($r_s = 0.0989$, RMSE $= 2.6199$). Within-Cell Drug-Only achieves 0.8554 ± 0.0598 on Random and 0.8449 ± 0.0602 on Leave-Cell-Out.

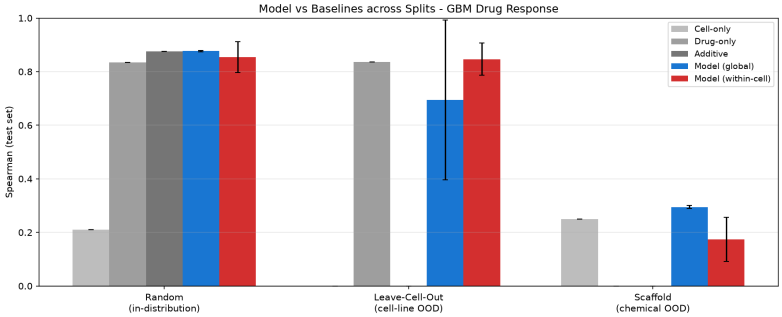


Figure 2: Comparison of statistical baselines (`Cell-only`, `Drug-only`, `Additive`) and `BaselineMLP` (global and within-cell Spearman) across Random, Leave-Cell-Out, and Scaffold splits. Error bars denote ±1 standard deviation across 5 random seeds.

Training procedure and Software 3.0 engineering contributions. Target LN_IC50 values are Z-score standardized using training statistics (μ_y, σ_y) and trained under MSE loss using Adam ($\eta = 10^{-4}$, weight decay $\lambda = 10^{-4}$, batch size 64, dropout 0.30 in `src/mlp_training.py`) across 5 seeds (seed $\in \{0, \ldots, 4\}$) for up to 100 epochs with early stopping patience 15 on validation RMSE. Predictions are unstandardized via $\hat{y}_{d,c} = \sigma_y \hat{z}_{d,c} + \mu_y$. Using DeepSeek (for modular system architecture design) and Gemini (for code implementation), we refactored our exploratory notebooks into a two-layer package of 15 atomic modules and 5 workflow orchestrators (`src/`). Human verification resolved two AI-generated bugs during refactoring: dynamically inferring $d_{\text{cell}} \in \{40, 44\}$ rather than hardcoding 44, and adding cell-line stratification in `partition_train_validation`.

5 Experiments

5.1 Evaluation metrics and RQ1: Shortcut decomposition on the Random split

We report four metric groups: (1) **Global Spearman (r_s) and RMSE** across all test pairs (5-seed mean ± SD); (2) **Within-Cell Spearman** across drugs per cell line ($\geq$ 10 drugs); (3) **Median Within-Drug Spearman** across cell lines per drug ($\geq$ 10 cells on Scaffold, $\geq$ 3 on Random/Leave-Cell); and (4) **Prediction variance shrinkage** $\sigma(\hat{y})/\sigma(y)$, mean signed bias, and large-error fraction $|\hat{y} - y| > 2.0$ ln μM (`src/uncertainty.py`).

Table 2 and Figure 2 answer **RQ1**. On the Random split (1,022 test pairs), `BaselineMLP` achieves a global Spearman of 0.8744 ± 0.0051 (RMSE 1.0346 ± 0.0211). However, predicting only the marginal cell-line mean (`Cell-Only`) yields 0.2107, whereas predicting only the marginal training drug mean (`Drug-Only`, ignoring all patient omics) yields 0.8349—explaining 95.5% of the deep network's rank correlation. Adding marginal drug and cell means without interaction (`Additive` baseline, Eq. 1) achieves 0.8752, matching `BaselineMLP` (0.8744, net interaction gain $\Delta r_s = -0.0008 \approx 0$). Within individual cell lines, ranking drugs by their training mean (`Within-Cell Drug-Only` $= 0.8554 \pm 0.0598$) also matches `BaselineMLP` (0.8541 ± 0.0576, $\Delta r_s = -0.0013$). Thus, high random-split correlation is overwhelmingly driven by memorizing per-drug average toxicity.

Table 3: Prediction variance shrinkage ($\sigma(\hat{y})/\sigma(y)$), signed bias (mean($\hat{y} - y$)), large-error rate ($|\hat{y} - y| > 2.0$ ln μM), and per-drug median metrics across splits.

Split Regime	Pred SD $\sigma(\hat{y})$	True SD $\sigma(y)$	Shrinkage $\sigma(\hat{y})/\sigma(y)$	Mean Bias	Frac. $\lvert e\rvert > 2.0$	Per-Drug Med. RMSE	Within-Drug Med. r_s
Random (In-Dist.)	2.267	2.631	0.862	−0.055	6.0%	0.893	0.500
Leave-Cell-Out (OOD)	2.209	2.592	0.852	−0.147	7.0%	0.887	0.500
Scaffold (Chem. OOD)	1.630	2.459	**0.663**	−0.316	**36.8%**	**1.644**	**0.530**

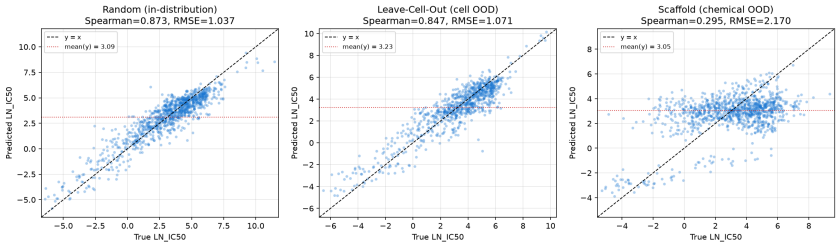


Figure 3: Predicted vs. true LN_IC50 scatter plots across Random (left), Leave-Cell-Out (middle), and Bemis–Murcko Scaffold (right) test sets. On the Scaffold split, unseen chemical structures cause predictions to compress horizontally toward the global training mean ($\bar{y} \approx 2.85$–3.05, red dotted line), reducing $\sigma(\hat{y})/\sigma(y)$ from 0.862 to 0.663.

5.2 RQ2: Biological OOD vs. Chemical OOD and variance shrinkage

Biological OOD (Leave-Cell-Out) and multi-seed stability. On Leave-Cell-Out (4 unseen GBM cell lines, 858 pairs), `Drug-Only` achieves 0.8362 globally and 0.8449 ± 0.0602 within-cell because drug potency ordering is conserved across GBM cell lines. Four of five neural network seeds (seed $\in \{0, 1, 3, 4\}$) converge stably to $r_s = 0.8436 \pm 0.0034$ (RMSE 1.0912 ± 0.0085), whereas seed $= 2$ early-stops at epoch 21 with $r_s = 0.0989$ (RMSE 2.6199), lowering the 5-seed mean to 0.6947 ± 0.2979 (Table 2). With only 20 training cell lines, validation loss on a small subset can plateau prematurely if initial cell-tower weights are poorly aligned, demonstrating why multi-seed auditing (`src/uncertainty.py`) is essential.

Chemical OOD (Scaffold split) and variance shrinkage. On the Scaffold split (47 unseen drugs, 1,051 pairs), drug-identity lookup is impossible (`Drug-Only` coverage $= 0\%$). Global Spearman drops from 0.8744 to 0.2905 ± 0.0045, Within-Cell Spearman drops to 0.1735 ± 0.0813, and RMSE doubles from 1.0346 to 2.1951 ± 0.0081 (though `BaselineMLP` still doubles the correlation of `Nearest-Drug Morgan FP` at 0.1445). As shown in Table 3 and Figure 3, unfamiliar Morgan fingerprints cause drug-tower outputs to regress toward the global training mean under MSE loss: the shrinkage ratio $\sigma(\hat{y})/\sigma(y)$ contracts from 0.862 (Random) and 0.852 (Leave-Cell-Out) to 0.663 (Scaffold), and the fraction of test predictions with $|\hat{y} - y| > 2.0$ ln μM jumps from 6.0% to 36.8%.

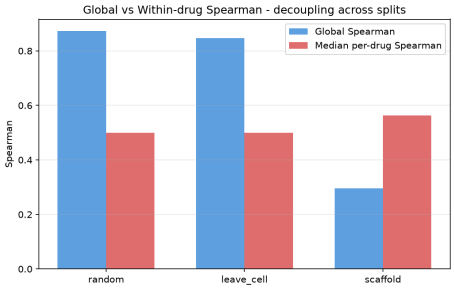
5.3 RQ3: Decoupling between global collapse and within-drug patient ranking

Comparing global pooled correlation against within-drug correlation across cell lines reveals our central technical finding (Table 3 and Figure 4, left): while global Spearman drops from 0.8744 to 0.2905 on the Scaffold split, the **median within-drug Spearman across the 24 GBM cell lines remains** 0.530 (compared to 0.500 on Random and 0.500 on Leave-Cell-Out).

Mathematically, for a fixed unseen drug d_0, the drug embedding $\mathbf{h}_{\text{drug}}(d_0)$ is constant across all 24 GBM cell lines. Misestimating the baseline potency of d_0 adds a constant offset Δ_{d_0} to every cell line's prediction for that compound. Because rank ordering within drug d_0 is invariant to a constant shift Δ_{d_0}, the cell tower—which was trained on all 24 cell lines—still ranks relative patient susceptibility from multi-omics profiles ($r_s = 0.530$). When all 47 unseen drugs are pooled together, however, the uncalibrated offsets $\{\Delta_d\}$ corrupt cross-drug rankings and collapse global Spearman (0.2905).

5.4 Clinical GBM drug audit, scaffold novelty, and Software 3.0 testing

Table 4a shows the impact of variance shrinkage on clinical GBM drugs. For Temozolomide (TMZ), Random-split predictions are well-calibrated (true +6.45, pred +6.17, RMSE 0.50), but on the Scaffold split ($S_{\max} = 0.284$), predictions shrink toward the global mean (~ 2.85) to +3.32, causing



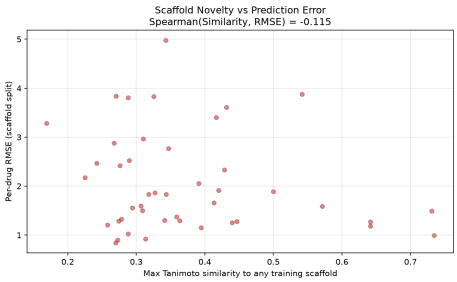


Figure 4: *Left:* Decoupling between global pooled Spearman (blue) and median within-drug Spearman across GBM cell lines (red) across splits. *Right:* Per-drug RMSE on the Scaffold split vs. maximum Tanimoto similarity to any training scaffold ($r_s = -0.115$).

Table 4: Audit of four clinical GBM chemotherapeutics across splits (`inspect_key_gbm_drugs`) and scaffold novelty RMSE bins (`analyze_scaffold_novelty_vs_error`).

(a) Clinical GBM Drugs Audit Across Splits

Drug Name	Split	n	True Mean	Pred Mean	Bias / RMSE
Temozolomide	Random	3	+6.45	+6.17	−0.28 / 0.50
Temozolomide	Scaffold	22	+6.45	+3.32	**−3.13 / 3.23**
Irinotecan	Random	4	+2.76	+2.69	−0.07 / 0.72
Irinotecan	Scaffold	21	+2.76	−1.01	**−3.78 / 3.92**
Carmustine	Leave-Cell	1	+5.63	+5.15	−0.49 / 0.87
Etoposide	Random	4	+2.24	+1.88	−0.36 / 0.90

(b) Scaffold Split Tanimoto Similarity vs. RMSE

Max Train Tanimoto	Drugs	Mean RMSE	Median RMSE
< 0.30 (High novelty)	15	2.112	**2.161**
0.30–0.50 (Moderate)	23	2.120	1.871
0.50–0.70 (Near-analog)	4	2.020	1.463
> 0.70 (Close scaffold)	2	1.253	**1.253**
Continuous Spearman($S_{\max}$, RMSE) $= -0.115$			
(72% higher median RMSE for < 0.30 vs. > 0.70)			

a -3.13 log-unit bias (RMSE 3.23). Likewise, Irinotecan ($S_{\max} = 0.241$) shifts from RMSE 0.72 on Random to -3.78 log bias (RMSE 3.92) on Scaffold. Across all held-out drugs (Table 4b and Figure 4, right), highly novel scaffolds ($S_{\max} < 0.30$) suffer a 72% higher median RMSE (2.161) than close analogs ($S_{\max} > 0.70$, median RMSE 1.253). Finally, Gemini assisted in building our 58-test unit suite (`tests/`), automated GitHub Actions CI (`.github/workflows/ci.yml`), single-file project showcase (`docs/index.html`), and interactive `LecTrace`[3] script (`Lectrace/01_gbm_ood_presentation.py`), which traces (1) train-only standardization, (2) a forward pass showing a $+1.258$ ln μM resistance shift between `MGMT`-low `U-87 MG` and `MGMT`-high `T98G` on TMZ, (3) additive shortcut decomposition, and (4) chemical OOD shrinkage.

6 Conclusion

We presented a leakage-free multimodal benchmark auditing shortcut learning and out-of-distribution generalization across 5,107 GDSC2 and DepMap Glioblastoma drug–cell pairs (24 GBM cell lines and 243 small-molecule drugs). Our study yields three main findings for precision oncology machine learning:

1. **In-distribution correlation is dominated by drug-mean memorization:** On the Random split, a closed-form zero-interaction `Additive` baseline ($r_s = 0.8752$) matches our Two-Tower `BaselineMLP` (0.8744 ± 0.0051), with 95.5% of explainable rank correlation driven by per-drug average toxicity (`Drug-Only` $r_s = 0.8349$). Preclinical drug response benchmarks should report additive baselines alongside neural network metrics to rule out shortcut learning.
2. **Chemical OOD shift induces severe prediction variance shrinkage:** While biological OOD (Leave-Cell-Out) preserves high correlation across stable seeds (0.8436), holding out unseen Bemis–Murcko chemical scaffolds reduces global Spearman to 0.2905 ± 0.0045 and compresses prediction standard deviation ($\sigma(\hat{y})/\sigma(y) = 0.663$), causing large systematic potency offsets on clinical GBM agents such as Temozolomide (-3.13 log units) and Irinotecan (-3.78 log units).
3. **Relative patient stratification survives chemical OOD:** Despite global collapse on unseen scaffolds, median within-drug Spearman across the 24 GBM cell lines remains 0.530. Thus, while current multi-omics neural networks must not be used to select uncalibrated clinical

[3]LecTrace interactive execution-explanation framework by Mr. Adesanmi Dayo Praisegod: https://pypi.org/project/lectrace/.

dosages for novel compounds, their cell-line encoders retain signal for ranking relative patient susceptibility in biomarker-stratified oncology trials.

Limitations and future work. First, GDSC2 assays measure 72-hour in-vitro viability in 2D immortalized cell lines, which do not capture blood–brain barrier permeability, tumor microenvironment interactions, or systemic toxicity in patients. Second, our cohort is limited to 24 GBM cell lines; expanding to patient-derived glioma stem-like cells (GSCs) will improve statistical power for Leave-Cell-Out evaluation. Future work will investigate pre-trained 3D molecular foundation encoders to mitigate chemical OOD shrinkage and few-shot single-concentration calibration to anchor baseline potency offsets for novel scaffolds.

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